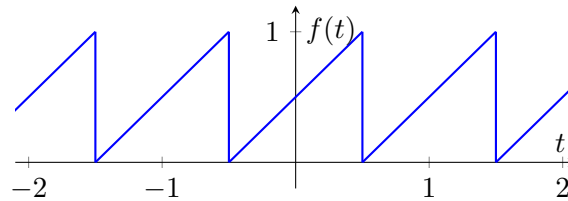


ECE 3210 quiz03

Name: _____ SOLUTIONS

Consider the periodic signal $f(t)$ as seen in the figure below.



1. (1 point) What is the period of $f(t)$? What is the fundamental frequency of $f(t)$?

Solution: The period of $f(t)$ is $T = 1$. The fundamental frequency of $f(t)$ is $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{1} = 2\pi$.

2. (8 points) Find the complex exponential Fourier series representation of $f(t)$.

Solution: Let's make this easier. First, let's delay the signal by $T/2$ so that $\tilde{f}(t) = t$ for $0 < t < 1$. Then we can find the Fourier series coefficients of $\tilde{f}(t)$ and then multiply by $e^{-j\omega_0 t_0}$ to get the coefficients of $f(t)$. The Fourier series coefficients are given by

$$\begin{aligned}\tilde{D}_n &= \frac{1}{T} \int_0^T \tilde{f}(t) e^{-jn\omega_0 t} dt \\ &= \int_0^1 t e^{-j2\pi n t} dt \\ &= \left[\frac{t e^{-j2\pi n t}}{-j2\pi n} + \frac{e^{-j2\pi n t}}{4\pi^2 n^2} \right]_0^1 \\ &= \frac{1}{-j2\pi n} + \frac{1}{4\pi^2 n^2} - \frac{1}{4\pi^2 n^2} \\ &= \frac{1}{-j2\pi n}\end{aligned}$$

We will need to account for the $n = 0$ case, which is $\tilde{D}_0 = \int_0^1 t dt = \frac{1}{2}$. We can construct the Fourier series of $\tilde{f}(t)$ as

$$\tilde{f}(t) = \frac{1}{2} + \sum_{n=-\infty, n \neq 0}^{\infty} \frac{1}{-j2\pi n} e^{j2\pi n t}$$

We recall that $f(t) = \tilde{f}(t + 1/2)$. We can add the $1/2$ time shift to the Fourier series

$$f(t) = \frac{1}{2} + \sum_{n=-\infty, n \neq 0}^{\infty} \frac{1}{-j2\pi n} e^{j2\pi n(t+1/2)}.$$

We can break apart the exponential to get

$$f(t) = \frac{1}{2} + \sum_{n=-\infty, n \neq 0}^{\infty} \frac{1}{-j2\pi n} e^{j2\pi n t} e^{j2\pi n(1/2)}.$$

We can simplify the last exponential term, which is

$$e^{j2\pi n(1/2)} = e^{j\pi n} = (-1)^n.$$

Putting it all together, we have the Fourier series of $f(t)$ as

$$f(t) = \frac{1}{2} + \sum_{n=-\infty, n \neq 0}^{\infty} \frac{j(-1)^n}{2\pi n} e^{j2\pi n t}.$$

As an aside, we did a time shift of $1/2$, which is the equivalent of

$$D_n = \tilde{D}_n e^{-j\omega_0 t_0} = \tilde{D}_n e^{-j2\pi n(1/2)} = \tilde{D}_n e^{-j\pi n} = \tilde{D}_n (-1)^n.$$

This is consistent with the Fourier shift property discussed in class.